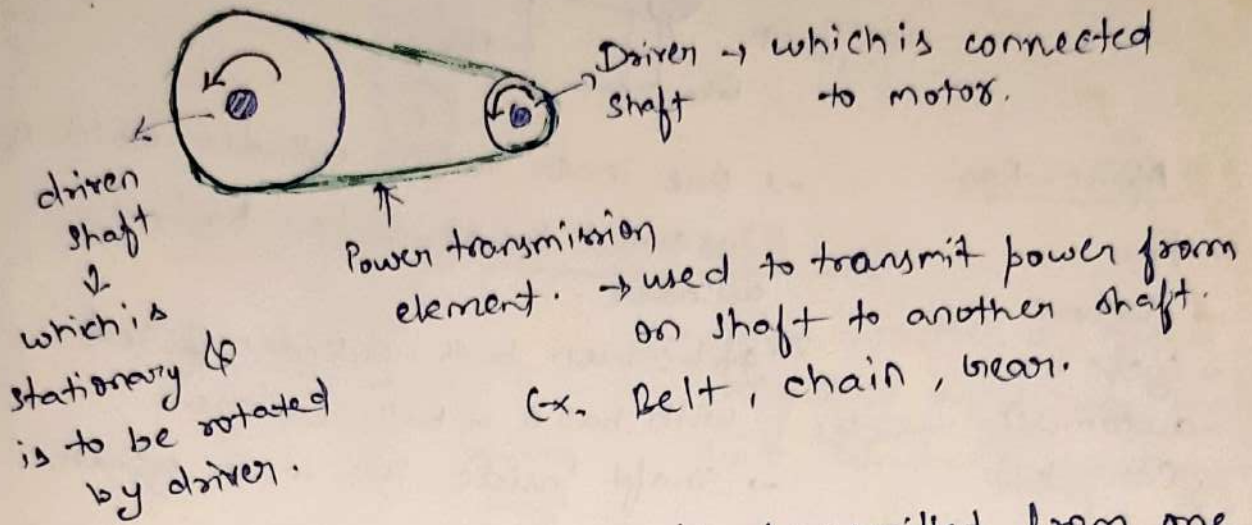


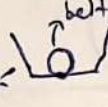
Power transmission system



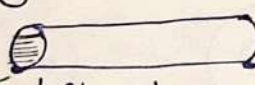
System in which power is transmitted from one driver shaft to driven shaft.

Different type of belt

(i) • Circular belt

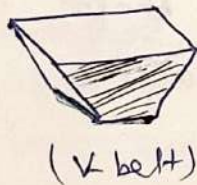
(rope belt) (pulley) 
 → It has point contact
 → So friction is less.
 Slip is more.

→ less power transmitted.
 efficiency less.

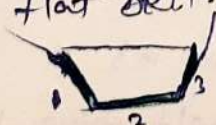
Point Contact (Circular) 

(iii) • V-belt

3 line contact
 friction is more.
 Slip ↓
 Power transmission is more.
 efficiency ↑ > flat belt.



(V belt)



(ii) • Flat belt



→ a line contact
 → friction is more
 → Slip is less.
 → more power transmitted.
 efficiency is more than Circular.



(iii) • Ribbed belt

• Timing belt

→ Internal surface is constructed by slots
 → Surface area contact is max.
 → friction max, Slip min.
 → Power transmission max.
 efficiency max.



* Ball Bearing



(9)

Application

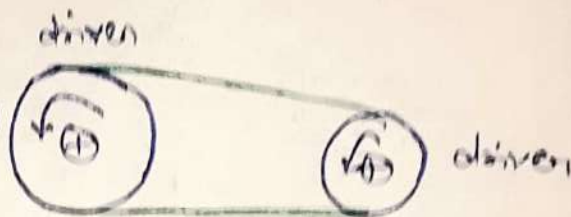
- Fan
- Mixture
- Cycle
- automobile vehicles
- Cars etc

- One inner hollow cylinder which is inserted inside outer hollow cylinder
- gap between both cylinder filled with balls → ball bearings.
- Shaft inside the inner cylinder

* Different type of belt Drive

(1) Open belt drive -

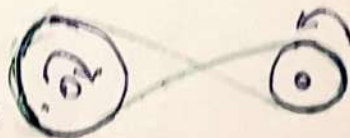
- One endless belt is mounted between driven & driven shaft.



- Power transmission in same direction, in which driven shaft is rotating.
- Used when axis of both shafts are parallel.

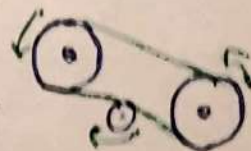
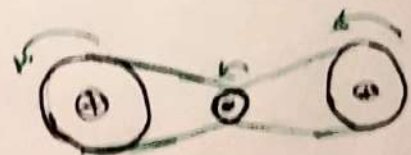
(2) Crossed belt drive -

- Similar to open belt drive.
- power transmission in opposite direction in which driven shaft is rotating.
- Used when axis are parallel.

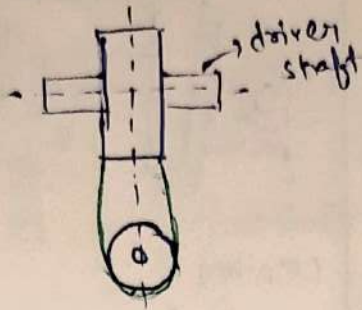


(3) Intermediate belt drive

- Used when 2 shafts are placed at large distance.
- One more pulley is placed between 2 shaft.
- Used for Id is also known as Idler Pulley belt drive. (Jockey)

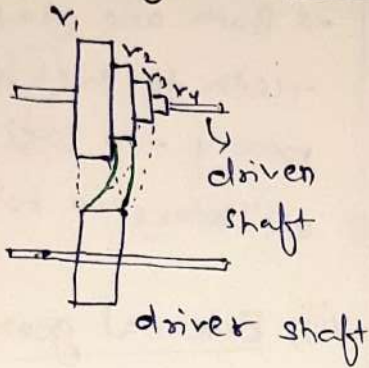


④ Quarterly (Right angled) belt drive - (2)



two shafts are arranged at right angle.

(5) Cone Pulley / Step Pulley belt drive -



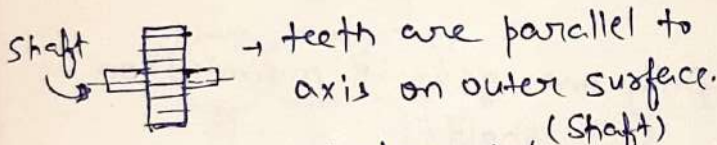
→ When varieties of speed is required on driven shaft with constant speed motor (driver)

→ It has more than one pulley (Step)

→ Shape $\Rightarrow$ Cone like

• Gears :- Power transmission element.

Types (i) Spur Gear



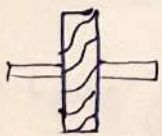
→ teeth are parallel to axis on outer surface. (Shaft)

→ Used in wrist watch.

→ vernier caliper (Precision measuring instrum.)

→ for low power transmission.

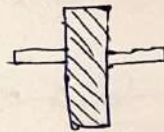
(ii) Spiral Gear



→ teeth are cut in spiral form on outer surface.

→ when axis are not parallel but intersecting each other.

(iii) Helical Gear

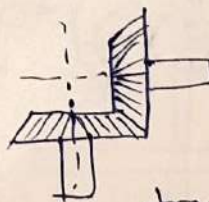


teeth are cut in helical form

→ Used for high speed.

→ Smooth operation.

(iv) Bevel gear

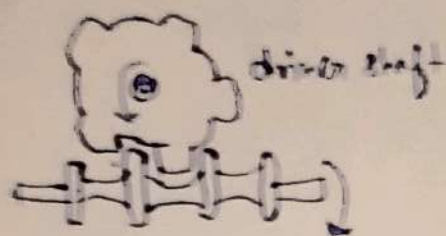


It has two cone gear.

It is used to transmit power at an angle 90° .

It is used in automobile.

(vi) Worm gear



driven shaft.

→ It has a large wheel on which teeth are constructed.

→ below that wheel, screw is placed.

→ when wheel will rotate, screw will also rotate.

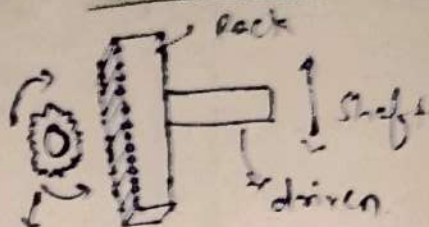
at 90° or 180°

(vii) Internal gear



Teeth are inside of gear ring.

(vi) Rack gear



Driver (pinion)

→ It has one small gear and one slab with teeth.

→ Both are meshed with teeth to each other and rotary → oscillatory
oscillatory → rotary motion.

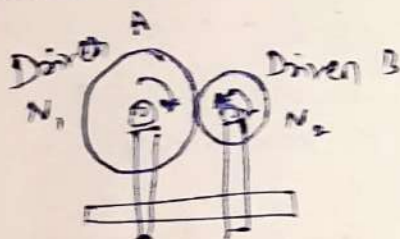
(viii) External gear



teeth are outside of gear ring.

* Gear train

(i) Simple gear train → Only one gear is mounted on each shaft.



$$\text{Train value} = \frac{N_2}{N_1} = \frac{\text{Driven speed}}{\text{Driver speed}}$$

Note $\frac{D_A}{D_B} = \frac{N_2}{N_1}$

is No of (Intermediate gear)

is even

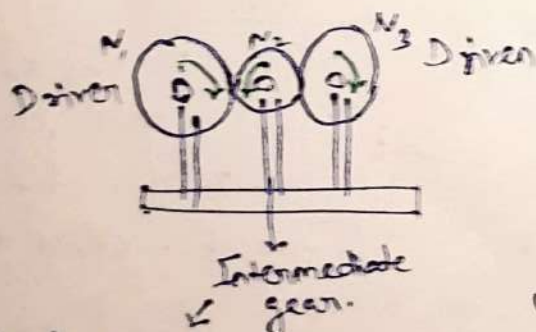
then direction of speed of driver & driven are opposite.

(ii) No of (Intermediate gear) is odd.

then direction of speed is same

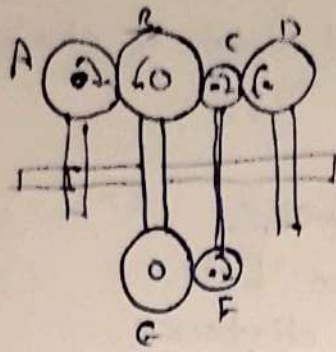
Used when to cover the gap between driver &

driven. as desired direction of velocity.

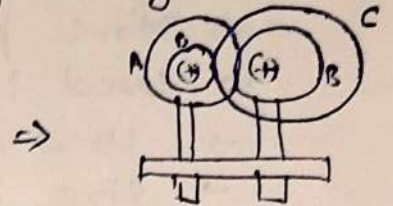
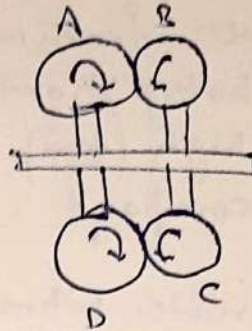


(ii) Compound gear train:-

5



When two or more gear are on one shaft & rpm of both shaft is same, then this type of arrangement is known as compound gear train.



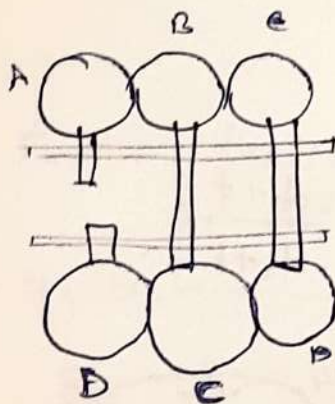
$$r_A + r_B = r_D + r_C$$

Speed ratio

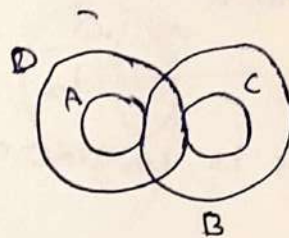
$$\frac{N_A}{N_B} = \frac{D_B}{D_A} \quad \& \quad \frac{N_C}{N_D} = \frac{D_D}{D_C}$$

(iii) Reverted gear train:-

When driver & driven are mounted on ~~same~~ shaft, co-axial shaft.



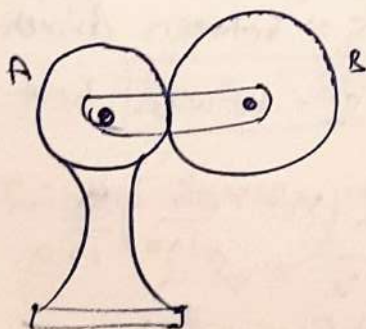
=>



$$r_A + r_B = r_C + r_D$$

(iv) Epicyclic gear train:-

When one gear rotates in cyclic motion about another gear.



→ The axes of shaft, over which the gear more mounted, may move relative to a fixed axis.

* Chain Drive

(6)

→ In belt drive some slip is present, but in chain drive, it is eliminated.

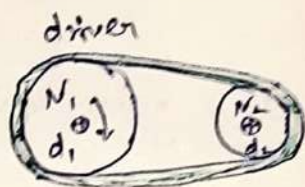
Advantage ⇒ No Slip, Accurate speed.

- More power transmission than belt.
- Used in both short & long distance.
- Used at high temp as well as low temp.
- Size is compact.

Disadvantage ⇒ → Creates vibration

- Production cost is high.
- Proper alignment is required.
- Required maintenance & lubrication.

* Velocity Ratio = $\frac{N_{\text{driver}}}{N_{\text{follower}}}$

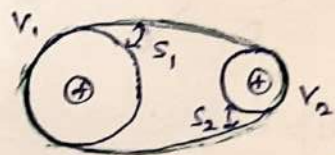


$$V_{\text{driver}} = \pi d_1 N_1$$

$$V_{\text{follower}} = \pi d_2 N_2 \Rightarrow \frac{d_1}{d_2} = \frac{N_2}{N_1}$$

if thickness is considered ⇒ $\boxed{\frac{N_2}{N_1} = \frac{d_1 + t}{d_2 + t}}$

* Slip = some forward motion of the driver without carrying the belt with it (relative motion)



$$(\text{Speed})_{\text{belt}} = V_1 - V_1 \times \frac{s_1}{100}$$

s_1 ⇒ between driver & belt
 s_2 ⇒ between belt & driven

driven speed

$$V_2 = \left(V_1 - V_1 \frac{s_1}{100} \right) - \left(V_1 - V_1 \frac{s_1}{100} \right) \frac{s_2}{100}$$

$$N_2 = \frac{V_1 (100 - s_1) (100 - s_2)}{10000}$$

$$\therefore V = \frac{\pi d N}{60}$$

$$\Rightarrow \frac{\pi d_2 N_2}{60} = \frac{\pi d_1 N_1}{60} \times \frac{(100 - s_1)(100 - s_2)}{10000}$$

$$\Rightarrow \frac{N_2}{N_1} = \frac{d_1}{d_2} \left[1 - \left(\frac{s_1 + s_2}{100} \right) \right] \quad \left(\because s_1 + s_2 \approx S \text{ total slip} \right)$$

$s_1, s_2 \approx 0$

$$\boxed{\frac{N_2}{N_1} = \frac{d_1}{d_2} \left(1 - \frac{S}{100} \right)}$$

* Creep of Belt :- When the belt passes from slack side to tight side, a certain portion of belt extends. (also when from tight to slack side)

Due to length change, relative motion occurs known as Creep

effective turning force = $T_1 - T_2$

$$\text{work done/sec} = P = (T_1 - T_2)v$$

Friction force

$$F = \mu R_N$$

By NLM,

$$R_N = T \sin \frac{\delta\theta}{2} + (T + \delta T) \sin \frac{\delta\theta}{2} \quad (\text{horizontal})$$

$$R_N = T \frac{\delta\theta}{2} + \cancel{\delta T \frac{\delta\theta}{2}} + \frac{T \delta\theta}{2} \quad (\delta\theta \approx \text{small})$$

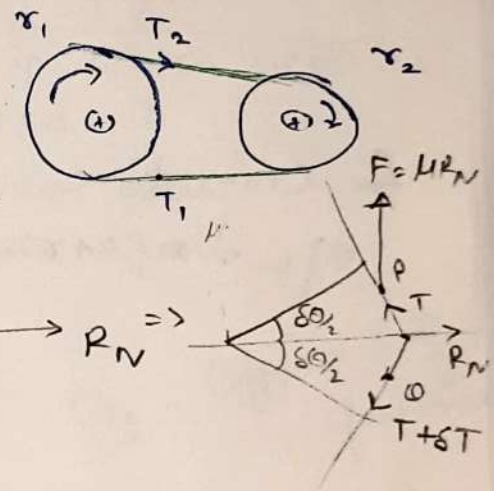
$$\boxed{R_N = T \delta\theta}$$

Friction force in vertical direction,

$$\mu R_N = (T + \delta T) \cos \frac{\delta\theta}{2} - T \cos \frac{\delta\theta}{2}$$

$$\mu R_N = \delta T$$

$$\boxed{R_N = \delta T / \mu}$$



$$\frac{\delta I}{T} = \mu \theta \quad (T_1 \text{ to } T_2)$$

$$\Rightarrow \left[\frac{T_1}{T_2} = e^{\mu \theta} \right] \quad (\text{Tension Ratio})$$

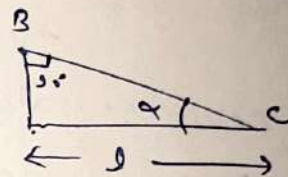
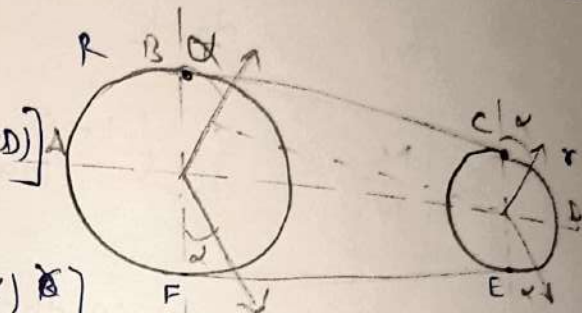
* Length of Belt
(Open drive)

$$\text{Length} = 2 [\text{Arc}(AB) + BC + \text{Arc}(CD)]$$

$$= 2 \left[\left(\frac{\pi}{2} + \alpha \right) R + BC + \left(\frac{\pi}{2} - \alpha \right) r \right]$$

$$= 2 \left[\pi(R+r) + 2\alpha(R-r) - \left(\frac{R-r}{1} \right)^2 + 2l \right]$$

$$L = \pi(R+r) + \frac{(R-r)^2}{1} + 2l$$



$$BC = 1 \left[1 - \left(\frac{R-r}{2l} \right)^2 \right]$$

for cross drive

$$\Rightarrow L = \pi(R+r) + \frac{(R+r)^2}{1} + 2l$$

$$\sin \alpha = \frac{R-r}{l}$$

$$(\alpha = \frac{R-r}{l})$$